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CHAPTER 4

Regional identity in Highland Ecuador

Social evaluation of intervocalic /s/ voicing

Christina García
Saint Louis University

Of considerable interest to linguists recently is the variable voicing of intervocalic /s/, which has been attested in dialects as diverse as Ecuadorian (Chappell, 2011; García, 2015; among others) and Catalonian Spanish (Davidson, 2014; McKinnon, 2012), among others. While our knowledge of the production of this variable has advanced, the question remains of how /s/ voicing is evaluated socially (although see Chappell, 2016). This chapter details a matched-guise experiment comparing the evaluation of intervocalic [s] and [z] in one coastal and three highland Ecuadorian cities. The results show that this variable is a regional marker within the highlands, and that it is also associated with status, pleasantness, and age, but only for female speakers. In contrast to other studies on social meaning, it is only females' use of intervocalic [z] that is socially-charged, whereas males' use is not strongly associated with any of the social characteristics examined.

1. Introduction

Research in the past few decades in the area of sociolinguistic perception has sought to disentangle the ways in which social and linguistic information contribute to speaker perception. These studies largely indicate a bidirectional relationship between the two, in which the linguistic signal influences social perception, and any social information available to the listener conditions what is heard or how it is perceived (see Campbell-Kibler, 2010). For example, Walker, García, Cortés, and Campbell-Kibler (2014) show that both speaker and listeners' origin influences the social evaluation of /s/ aspiration in Spanish. Additionally, Chappell (2019) finds that hearing sibilant or aspirated /s/ affects whether a given speaker is heard as Mexican or Puerto Rican. At the same time, scholars working in perceptual dialectology have found that dialect categorization is dependent on many signal and listener variables. For instance, Clopper and Pisoni (2004) contend

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that speaker evaluations are formed based on a small set of salient variables, and also show that listeners are generally more accurate at categorizing variation that is present in their own dialect, something that is also corroborated by Schmidt (2013). Furthermore, a large body of work in this field suggests that listeners tend to favor their own variety over others, except in the case of linguistic insecurity (see Preston, 1999, for an overview).

Building on Walker et al. (2014) and Chappell (2019), the present study looks at the intersection of these two related fields (sociolinguistic perception and perceptual dialectology) by comparing the language attitudes of listeners from four Ecuadorian cities. The language attitudes examined here concern intervocalic /s/ voicing, a variable process in which intervocalic /s/ in word-initial (*la sopa* [la. ˈzo.pa] 'the soup'), word-medial (*casa* [ˈka.za] 'house'), or word-final (*las alas* [la. ˈza.las] 'the wings') position is realized as [z]. Using the matched-guise technique (Lambert, Hodgson, Gardner, & Fillenbaum, 1960), this variable is manipulated in the speech of two male and two female speakers from Loja, Ecuador, and ratings are compared between the [s] and [z] guises. I show that the social meanings of intervocalic [z] in Ecuadorian Spanish are somewhat more limited than those of Costa Rican Spanish (Chappell, 2016), and that gender and regional origin intersect in interesting ways in the evaluation of this variable.

2. Literature review

2.1 Ecuadorian Spanish

Scholars working on Ecuadorian Spanish generally agree that this macrodialect can be broken up into three major sub-dialects: Coast, Highlands, and Amazon (Toscano Mateus, 1953; Boyd-Bowman, 1953; Lipski, 1994; among others), observable in the map in Figure 1. Although the present study focuses on phonetic variation, there are differences between these sub-dialects at all linguistic levels. For instance, Highland Ecuadorian Spanish (HES) is easily distinguished from Coastal Ecuadorian Spanish (CES) due to the lexical and grammatical influence of Kichwa and other indigenous languages (see Haboud, 1998). Besides /s/, which is discussed in the next section, the production of /ʝ/-/ʎ/ and /r/ are the most salient phonetic features that distinguish HES and CES. On the one hand, CES is typically described as exhibiting *yeísmo* (/ʝ/ and /ʎ/ have merged to one phoneme) and a normative alveolar trill /r/, whereas *lleísmo* (maintenance of /ʝ/ and /ʎ/ as separate phonemes) and assibilated /r/ are variably present in HES. For the Amazonian region, Lipski (1994) affirms that a distinct "Amazonian Ecuadorian Spanish" has not yet emerged because most of its population are transplants from adjacent

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highland provinces, although this claim merits further investigation since it was made over twenty years ago. [1]

Figure 1. Map of Ecuador with linguistic regions according to Lipski (1994)

HES is generally further divided into smaller sub-dialects, as seen in Figure 1. Gómez (2003) and Lipski (1994) agree that the delateralization of /ʎ/ and assibilation of /r/ serve as isoglosses, demarcating Loja, Cañar/Azuay (typified by the city of Cuenca), and the Central Highlands (from Quito to the border of Cañar). Lojano and Cuencano traditionally exhibit palatal lateral /ʎ/, which is realized as a pre-palatal sibilant fricative in the Central Highlands (*malla* ['ma.ʒa] 'net'). As for /r/, it is realized as a normative trill in Lojano and assibilated in Cuencano and the Central Highlands, although to different degrees. As will be seen, these are the three regions that have been differentiated in the production of intervocalic /s/ as well.

2.2 Intervocalic /s/ voicing

In modern dialects of Spanish, /s/ is arguably the sound that varies the most and is most studied by linguists. There are two main threads of research pertaining to

1. Adapted from <http://www.luventicus.org/mapas/ecuador/esmeraldas.html>.

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Spanish /s/: weakening[2] and voicing. The former, the variable production of /s/ as [h] (aspiration) or a phonetic zero (ø), is dependent on various linguistic and social factors, and in Ecuador, is only found systematically in CES (King, 1953). The latter can be split into studies that examine the pre-consonantal assimilation of /s/ (Schmidt & Willis, 2011; A. García, 2013; Campos-Astorkiza, 2014; among others) and those that concern the voicing of intervocalic /s/. The variable voicing of intervocalic /s/ has been documented in the Spanish of the following places: Catalonia (McKinnon, 2012; Davidson, 2014), Madrid (Torreira & Ernestus, 2012; Hualde & Prieto, 2014), Central Spain (Torreblanca, 1986), Northern Spain (Bárkányi, 2014), Highland Ecuador (Calle, 2010; Chappell, 2011; García, 2015; Lipski, 1989; Robinson, 1979; Schmidt, 2016; Strycharczuk, Van'T Veer, Bruil, & Linke, 2014), and Costa Rica (Chappell & García, 2017).

The realization of intervocalic /s/ varies greatly between dialects, as do the conditioning factors. The linguistic factor most consistently found important is position within the word: the majority of studies find highest rates of voicing in word-final context (Robinson, 1979; Lipski, 1989; Calle, 2010; Chappell, 2011; Torreira & Ernestus, 2012; McKinnon, 2012; Strycharczuk et al., 2014; Davidson, 2014; Schmidt, 2016; Chappell & García, 2017), although Hualde and Prieto (2014) and García (2015) find most voicing in word-final *and* word-initial contexts. Several studies concur that intervocalic /s/ voicing is a reduction process, since it is favored at higher speech rates and between unstressed vowels (Torreira & Ernestus, 2012; García, 2015; Chappell & García, 2017). Results for word class and frequency are inconclusive, with some showing an effect on voicing (Chappell, 2011; Davidson, 2014), and others no effect (Torreira & Ernestus, 2012). For preceding/following segment, Chappell (2011) finds more voicing before a low vowel, while García (2015) finds more voicing before non-high vowels.

Social factors also condition this variable. Overwhelmingly, when gender is considered, studies find more voicing in males' speech (García, 2015; Schmidt, 2016; Chappell & García, 2017; among others), although McKinnon (2012) finds the opposite. Whether this is due to anatomical or social gender differences has been debated (see Chappell & García, 2017, for a complete discussion). Calle (2010) and García (2015) find opposite effects of speaker age on this variation: older speakers voicing more in the former and younger speakers voicing more in the latter. Many previous studies of intervocalic /s/ voicing also find considerable interspeaker variation. Finally, studies in Catalonia consider the effect of bilingualism (McKinnon, 2012; Davidson, 2014), exhibiting conflicting results as to whether bilingual, Catalan-dominant, or Spanish-dominant speakers voice the most.

2. "Weakening" has also been used to describe the voicing of intervocalic /s/, but this term here refers to the production of /s/ as [h] or (ø).

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Not only do rates of /s/ voicing differ between sub-dialects of HES, but the nature of the phenomenon does as well. Intervocalic /s/ voicing in Quiteño (Central Highlands) and Cuencano is generally characterized as categorical or *almost* categorical in word-final position (Robinson, 1979; Lipski, 1989; Chappell, 2011). Nevertheless, Strycharczuk et al. (2014) and Schmidt (2016) find that some Quiteño participants do not voice categorically. These studies agree, however, that there is negligible voicing in word-medial and initial contexts in Quiteño. Robinson (1979) adds that Cuencano also voices /s/ categorically at word-medial morpheme boundaries (e.g., *bi<u>s</u>abuelo* 'great-grandfather'). In comparison, the voicing in Lojano is a gradient, variable process, and considerable voicing is found in all three positions (García, 2015). Importantly, although not studied empirically, it is assumed that CES does not exhibit intervocalic /s/ voicing, because some of these cases of /s/ are variably aspirated in CES (King, 1953).

2.3 Perception of /s/

The "third wave" of sociolinguistics (Eckert, 2012) is increasingly concerned with not just what speakers produce, but what speakers *do* with language. Most relevant to the present study is the relationship between sounds and social meaning. It has been shown that several features contribute to the differentiation of dialects of Ecuadorian Spanish, but which of these features are socially salient to naïve native speakers remains unknown. For the perception of Spanish /s/, there are two main veins of previous research: those concerned with dialect/sound classification, and those investigating the social meaning of /s/.

In the first, studies show that previous linguistic experience influences how variation, specifically /s/ aspiration, is categorized. Boomershine (2006) demonstrates differences between Mexican and Puerto Rican participants, although both groups display slower processing for coda /s/ than other variables, due to the vast amount of variation present for coda /s/. Schmidt (2013) finds parallel results: Argentine participants (who are more exposed to /s/ aspiration) correctly identified stimuli with coda [h] as words containing /s/ almost twice as often as (highland) Colombian participants.

Similarly, dialect (and thus exposure) has an important influence on the social meaning of /s/ aspiration. Walker et al. (2014) show that listener and speaker dialect only affects some social dimensions: Mexican and Puerto Rican listeners rate Puerto Rican speakers using coda [h] less severely than Mexican speakers on measures of status; however, heteronormativity ('masculine-ness' and 'sexuality' combined) ratings do not exhibit dialect sensitivity. Their results for heteronormativity mirror those of Mack (2009): male speakers are heard as more masculine/heteronormative with coda [h] than sibilant [s]. These social meanings are further

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explored by Chappell (2019), who finds that whether [s] or [h] is heard can override other linguistic information in influencing where Mexican listeners think a speaker is from.

Few studies have looked specifically at the perception of intervocalic /s/ voicing. García (2015) and Chappell (2017) both compare the perception of [s]-[z] to other allophonic distinctions. The results of both studies suggest that Lojano and Costa Rican participants do perceive the [s]-[z] distinction, however, less consistently than other allophonic pairs. Chappell (2016) goes further to look at the social perception of intervocalic /s/ voicing in Costa Rica, finding that intervocalic [z] is associated with lower status for all speakers and also with solidarity measures for male speakers.

To date, no study has systematically looked at the social perception of intervocalic /s/ voicing in HES. Given the differences between /s/ voicing in HES and Costa Rican Spanish, it is likely that the social perception of this variable in HES is not identical to what Chappell (2016) finds. Most notably, /s/ voicing in Costa Rica is a gradient, phonetic process (Chappell & Garcia, 2017), while /s/ voicing in HES has been shown to be either gradient or categorical depending on region. Walker et al. (2014) find that some social meanings are more universal, while others are more local, which suggests that HES may share some of the associations found in Costa Rican Spanish, while also exhibiting others. The present study addresses the following research questions by investigating the perception of listeners from four regions of Ecuador:

1. What social characteristics are associated with intervocalic /s/ voicing in Ecuadorian Spanish?
2. Do these social meanings differ between the coastal and highland regions of Ecuador, and within the sub-dialects of the highlands?

3. Methodology

3.1 Stimuli

The stimuli for this experiment come from a corpus of sociolinguistic interviews collected in Loja (García, 2015). Speakers with considerable variation in intervocalic /s/ production were chosen so that there would be several options for splicing both [s] and [z] into other samples of their own speech, in order to create the stimuli. Two female and two male speakers were chosen, all natives of Loja, with ages ranging from 20 to 47 years (mean age: 30 years old). The stimuli material was taken from the portion of the interview in which the speakers were talking about their hometown, both for consistency between speakers and since speakers were theoretically most comfortable as it was the last portion of the interviews.

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This method was used as opposed to making new recordings in which the speakers are asked to purposefully produce [z] (cf. Walker et al. 2014) because it is unlikely that speakers would be able to reliably and naturally produce [z] on command.

From each speaker, two sentences were extracted that contained 4–6 tokens of intervocalic /s/, with at least one token in each of the three word positions: word-initial, word-medial, and word-final. I made two versions of each sentence: one with spliced[3] voiceless [s] and one with spliced voiced [z] for all tokens of intervocalic /s/. Tokens of /s/ in non-intervocalic contexts were not manipulated. The productions of [s] and [z] that were spliced in came from the same speaker, but a different section of the interview. The duration of the spliced /s/ was altered in some cases and a very small portion of the preceding/following vowel was also spliced in to improve naturalness. After splicing, the amplitude was equated across each stimulus so that the spliced portion would not stand out. These initial sixteen stimuli were checked impressionistically by two non-native trained linguists, one native trained linguist, and two naïve native Spanish speakers. Both naïve natives and linguists were used to ensure that any inconsistencies noted by linguists were not due to hyper-sensitive ears. Per speaker, the stimulus pair that was rated as most natural sounding and most easily distinguishable (as [s] or [z]) was chosen for use in the experiment, resulting in eight total target stimuli. A full list of the stimuli can be found in the Appendix. It is fair to assume that all stimuli are heard as representative of HES given the abundance of other HES features, such as lexicon and intonation. However, there was only one case of /r/ and no cases of /ɟ/ in the chosen stimuli. Since these two sounds are the main shibboleths differentiating sub-dialects of HES, it is assumed that the speakers could be interpreted as being from any part of the highland region.

3.2 Experiment design

The experiment was built and hosted online on Qualtrics (Provo, UT) and targeted native speakers of Ecuadorian Spanish from several regions. To ensure that participants would complete the whole survey, it was relatively brief. Upon consent, participants answered a series of basic demographic questions: their age, regional background, gender, which languages they speak, and whether they used speakers or headphones in the experiment. In addition to the eight target stimuli, there were eight fillers that serve as the targets for a different study.[4] I created two experimental lists of the stimuli (see Appendix) and each participant received one

3. The process of splicing involves delineating the original token of /s/, deleting it from the audio, and copying and pasting a different token of /s/ in its place.

4. The fillers were made from different speakers in the same corpus, but varied assibilated and alveolar trill /r/ instead of [s]-[z].

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of the two lists. These lists were designed so that participants heard one guise of each of the four speakers (e.g., [s]) before hearing the alternate guise (e.g., [z]) of any of the speakers. Within each list, the stimuli were pseudo-randomly organized to allow for maximal distance between the two guises of a particular speaker. The two lists differ in that List 1 presents Guise A of a given speaker in the first half whereas List 2 presents Guise B of that speaker first. This was done so that the first half of the results could be used if it were determined that the participants recognized the speaker repetition. The participants were told that they would listen to sixteen speakers and each stimulus was presented with a different name in hopes of distracting participants from the repetition.

Before the full data analysis was completed, the effect of list was examined and it was determined significant for two of the eight social characteristics rated. Additionally, 26 of the 219 participants explicitly commented on the repetition. For these reasons, I suspected that the second time they heard a given speaker, the participants were not rating that speaker based on the audio stimulus itself, but rather repeating the ratings they had already given that speaker. Accordingly, the dataset was halved so that only the first eight responses were used from each participant.

Following the design of Walker et al. (2014), participants were asked to evaluate each stimulus on a six-point scale for these social characteristics: *poco/muy femenina/masculino*[5] ('not very/very feminine/masculine'), *antipático/simpático* (roughly 'unpleasant/pleasant'), *inseguro/seguro de sí mismo* ('unsure/sure of himself/herself'), *de clase baja/de clase alta* ('low class/high class'), and *de pueblo/de ciudad* ('from a town/city'). An even number scale was used to avoid neutral responses. Participants were also asked to evaluate the speakers' origin from the following options: 'Quito,' 'Cuenca,' 'Loja,' and 'Other/I don't know.' These fixed options were given because these three highland cities represent the main sub-dialects of HES, as described in Section 2.1. Finally, participants were asked to evaluate the speakers' education level and age on five-point scales,[6] answer a free-response question regarding the speakers' ethnicity, and optionally leave additional thoughts about the speaker. On the final page of the survey, participants were able to leave any additional comments.

3.3 Participants

Participants were targeted in one coastal and three highland cities: Guayaquil, Quito, Cuenca, and Loja. Recruitment happened in person or through an email

5. '*Femenino*' appeared on the scale for female speakers and '*masculino*' for male speakers.

6. Five-point scales were used for these two qualities because there are not six potential age groups or education levels that would make sense for the population of interest.

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sent out to friends and colleagues in these four locations, asking them to forward to interested participants. This email specified that participants must be at least 18 years old, be a native speaker of Ecuadorian Spanish, and have not lived outside of Ecuador for more than three years, to ensure recent exposure to local language attitude norms. Any participant who did not meet these requirements or did not complete at least half of the survey was excluded from the analysis. Participants were compensated with a $5 cell phone credit upon survey completion.

A total of 219 people completed the survey. Table 1 presents a summary of the listeners' demographic information. As a result of the recruitment contacts, the vast majority of listeners are undergraduate students, but all have little or no linguistic training. Regional origin was asked as a free-response question to allow for maximally detailed information; however, the listeners were categorized into the four target regions after the fact. Listeners from towns or cities other than Guayaquil, Quito, Cuenca, and Loja were combined with the nearest category within the regional boundaries described in Section 2.1. All listeners from the coast were put into the same category for practical and linguistic reasons. The listeners responded to the language background question with the number of years they had studied English and I used this number to place them in four rough English proficiency categories: none, beginner, intermediate, and advanced. 'Beginner' is equivalent to 1–2 years of English instruction, 'Intermediate' to 3–5 years, and 'Advanced' to 6 or more years. English proficiency is examined given that /s/ and /z/ are phonemic categories in English, which could influence advanced learners' perception of these sounds. A few reported also having studied French, Kichwa, Portuguese, German, Italian, Chinese, Hebrew, and Romanian, most for fewer years than they have studied English, except in the case of two Spanish-Kichwa bilinguals.

Table 1. Listener demographics

	Listener's information
Median age (range), years	24 (18–64)
Male:female, *n*	92:127
Regional origin, *n*	Coast: 49, Quito: 39, Cuenca: 55, Loja: 76
English proficiency	none: 75, beginner: 35, intermediate: 65, advanced: 44
Total listeners	219

3.4 Statistical analysis

The data analysis follows that of Walker et al. (2014), Barnes (2015), and Chappell (2016). Responses on the scales were transformed to numerical values – 1 to 6 for masculinity/femininity, pleasantness, confidence, class, and urban-ness ratings,

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and 1 to 5 for education and age ratings. Perceived speaker origin was left as a categorical variable and perceived speaker ethnicity was not included in the analysis.[7] Ratings for femininity ('less feminine – very feminine') were inverted so that the scale would match those used for masculinity ('less masculine – very masculine'), assuming that 'less feminine' is equivalent to 'more masculine.' Since both six and five-point scales were used, all ratings were centered and standardized to make the scales comparable. In the results, zero should be understood as the central point, with positive numbers indicating more pleasant, older, etc., and negative numbers indicating less pleasant, younger, etc.

Since it seemed probable that some of the social characteristics are correlated, I performed factor analysis using the *factanal* function in R (R Core Team, 2014) with *varimax* rotation. Following Weatherholtz, Campbell-Kibler, and Jaeger (2014), 0.4 was used as the cutoff point in determining whether a given characteristic should be included in the aggregate factor. The results of the factor analysis suggest only one combined factor, corresponding to "status" (class, education, urban-ness, and confidence). The second factor is composed of only 'pleasantness,' the third factor of only 'age,' and 'masculinity/femininity' does not load onto any factor. Thus, these three factors were analyzed independently.

I modeled the aggregate and lone factors separately using mixed-effects linear regression with the *lmer* function (Bates, Maechler, Bolker, & Walker, 2014). The only categorical dependent predictor, perceived speaker origin, was modeled separately using multinomial logistic regression with the *multinom* function in the *nnet* package (Venables & Ripley, 2002). Participant was included in each model as a random effect,[8] along with the following independent variables:[9]

1. Variant (binary: [s] or [z])
2. Listener age (continuous)
3. Listener gender (binary: male or female)
4. Listener regional origin (categorical: Coast, Quito, Cuenca, Loja)[10]

7. Several participants commented that they did not know how to interpret the question about speakers' ethnicity. Additionally, ethnicity ratings are largely homogenous, with roughly 82% of responses being *mestizo/a* 'mixed race.'

8. Speaker was not included as a random effect since there are only four speakers and so few levels is not appropriate for a random effect (see, for example, the explanation in Gelman and Hill, 2006).

9. Listener gender and English knowledge were not selected as significant in any of the statistical models and thus are not mentioned in the results section.

10. These four categories were collapsed to Highland vs. Coast, but there were no significant differences between the analyses with four or two categories, so the more nuanced categories were used.

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5. Listener English knowledge (ordinal: none, beginner, intermediate, advanced)
6. Speaker gender (binary: male or female)
7. Perceived speaker origin (only for models where it is not the dependent variable; categorical: Quito, Cuenca, Loja, 'Other')

Treatment contrasts were used for categorical predictors and the corresponding reference levels are indicated in the results. In each regression model, independent variables were added one at a time in a stepwise procedure and nested models were compared using ANOVA, following Gries' (2013) approach of determining the "minimal adequate model." Interactions between main effects were tested and only significant interactions are retained in the models.

4. Results

The results of the statistical analyses show a significant effect of variant ([s] vs. [z]) in ratings of status, pleasantness, age, and perceived speaker origin, as well as effects of other independent variables. In the following tables, positive estimates for categorical variables indicate a higher rating as compared to the reference level (e.g., higher status), and negative estimates indicate a lower rating. In the case of perceived speaker origin (non-binary, categorical variable), the model with reference level as Loja is presented unless otherwise noted and results of re-leveling are mentioned where relevant. For continuous variables, positive estimates designate a positive correlation between the dependent and independent variables. The tables also include standard errors, *t* values, and *p* values (calculated from *t* values using the *lmerTest* package (Kuznetsova, Brockhoff, & Christensen, 2015)). The alpha value was set at 0.05. In the following boxplots, the diamond points denote the means for each group examined, while the line through each box represents the median. The box itself shows the middle 50% of all ratings, and the lines that extend above and below the box represent the highest and lowest 25% of all ratings, respectively.

4.1 Status factor

The best model for the combined status factor (class, education, urban-ness, and confidence) in Table 2 shows a main effect of variant, speaker gender, listener age, and perceived speaker origin, and an interaction between variant and speaker gender. There is an inverse relationship between listener age and status ratings such that younger listeners globally rate speakers as higher status than older listeners do. In comparison to speakers perceived as Lojano, speakers that are perceived

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Table 2. Summary of best mixed-effects model for status factor (N = 876)

	Estimate	*SE*	t value	*p*-value
(Intercept)	0.57	0.11	5.01	<0.001
Variant = [z]	−0.15	0.06	−2.30	0.02
Speaker gender = male	−0.60	0.07	−9.12	<0.001
Listener age	−0.009	0.004	−2.33	0.02
Perceived speaker origin = Cuenca	−0.17	0.08	−2.24	0.03
Perceived speaker origin = Quito	0.16	0.06	2.54	0.01
Perceived speaker origin = Other	−0.27	0.07	−3.97	<0.001
Variant = [z]: Speaker gender = male	0.21	0.09	2.20	0.03

as Cuencano or 'Other' receive significantly lower status ratings, while those perceived as Quiteño receive higher status ratings, independent of variant.

The more crucial results from Table 2 concern variant and speaker gender, whereby [z] is heard as lower status, and male speakers are heard as lower status than female speakers. The interaction shows, however, that variant only has a significant effect for female speakers, which is illustrated in Figure 2. The mean status rating for female speakers in the [s] guise is significantly higher than in the [z] guise. Post-hoc analysis (with Tukey correction) shows that the comparison between [s] and [z] is not significant for male speakers.

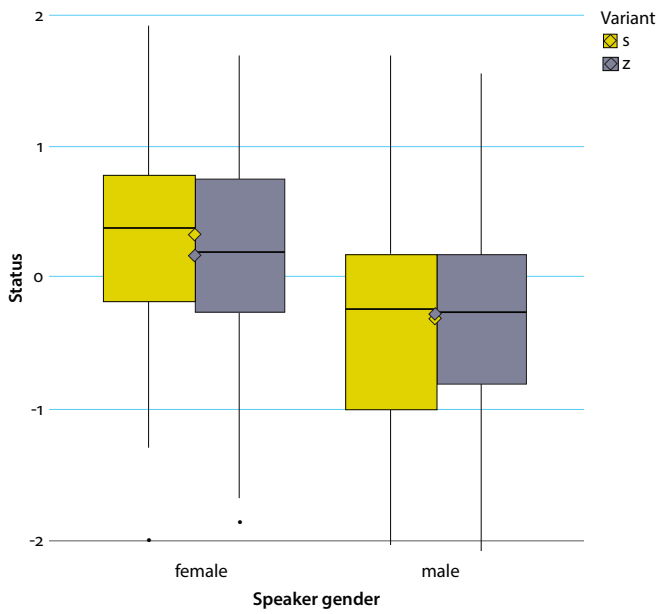

Figure 2. Boxplot of perceived status for [s] and [z] by speaker gender

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4.2 Pleasantness

Table 3 shows the best model for pleasantness ('*simpatía*') ratings, with significant main effects of variant and speaker gender, and an interaction between variant and speaker gender that is approaching significance.[11] Speakers are heard as less pleasant with the [z] variant than the [s] variant, and male speakers are rated as less pleasant than female speakers. Again, the interaction shows that the particular effect of variant differs between male and female speakers. As seen in Figure 3,

Table 3. Summary of best mixed-effects model for pleasantness ('*simpatía*') (N = 876)

	Estimate	*SE*	t value	*p*-value
(Intercept)	0.11	0.07	1.63	0.10
Variant = [z]	−0.28	0.09	−3.07	0.002
Speaker gender = male	−0.18	0.09	−2.05	0.04
Variant = [z]: Speaker gender = male	0.24	0.13	1.90	0.06

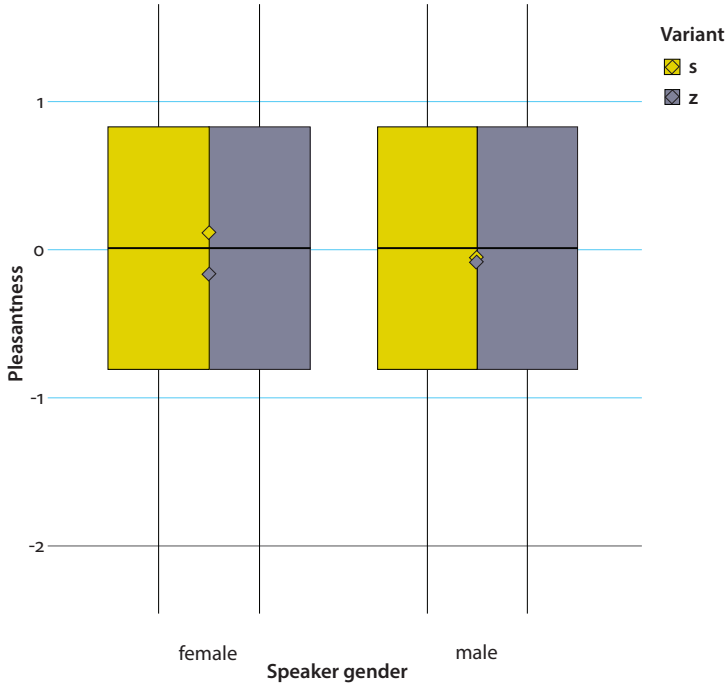

Figure 3. Boxplot of perceived pleasantness ('*simpatía*') for [s] and [z] by speaker gender

11. Although not significant, the interaction was left in the model because the effect of speaker gender is no longer significant when the interaction is taken out, and as seen in Figure 3, the interaction helps to illustrate the effect of variant.

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the mean pleasantness rating for females is significantly higher for the [s] guise than for the [z] guise, whereas the means for both variants are quite similar for male speakers, and post-hoc analysis (with Tukey correction) confirms the effect of variant is not significant for male speakers.

4.3 Age

The best model fit to perceived speaker age is shown in Table 4. This model includes main effects of variant, speaker gender, listener age and regional origin, perceived speaker origin, and the interaction between variant and speaker gender. A direct correlation exists between listener age and perceived speaker age in which older listeners hear speakers as significantly older than younger listeners do. Listeners from Cuenca and Loja hear speakers as significantly younger than listeners from Quito do. There is no significant difference in age ratings between listeners from the Coast and any other region. Perceived speaker origin also affects age ratings such that speakers perceived as Cuencano or Quiteño are heard as older than speakers perceived as Lojano.

Table 4. Summary of the best mixed-effects model for perceived speaker age (N = 876)

	Estimate	***SE***	**t value**	***p*-value**
(Intercept)	−0.29	0.18	−1.62	0.11
Variant = [z]	−0.31	0.09	−3.50	<0.001
Speaker gender = male	0.18	0.09	2.04	0.04
Listener age	0.01	0.005	2.79	0.005
Listener region = Cuenca	−0.23	0.10	−2.37	0.02
Listener region = Loja	−0.23	0.10	−2.32	0.02
Listener region = Coast	−0.08	0.11	−0.71	0.48
Perceived speaker origin = Cuenca	0.23	0.10	2.32	0.02
Perceived speaker origin = Quito	−0.19	0.08	−2.32	0.02
Perceived speaker origin = Other	−0.01	0.09	−0.12	0.92
Variant = [z]: Speaker gender = male	0.36	0.13	2.86	0.004

Most interesting are the effects of variant, speaker gender, and the interaction between the two. Overall speakers are rated as younger in the [z] guise as compared to the [s] guise, and male speakers are rated as older than female speakers. The interaction between these two factors is seen in Figure 4. With the interaction, it becomes clear that the effect of variant is only significant for female speakers, who are heard as younger in the [z] guise as opposed to the [s] guise. Post-hoc analysis (with Tukey correction) shows no significant effect of variant on age ratings for male speakers.

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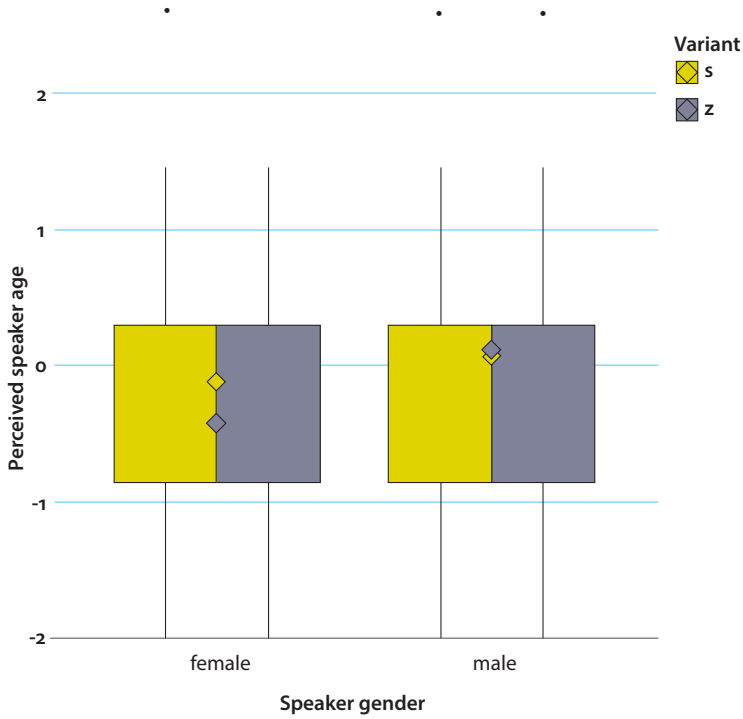

Figure 4. Boxplot of perceived speaker age for [s] and [z] by speaker gender

4.4 Masculinity/femininity

As an initial check, separate masculinity/femininity models were fit for female and male speakers, with the raw ratings before transformation. These models did not differ greatly from the best model with female and male speakers' data combined, and in none of the three models was variant a significant predictor. Thus, the results presented in Table 5 are those of the combined dataset, in which responses to female speakers have been inverted and all responses have been centered and scaled. These results show that male speakers are globally rated as more masculine than female speakers. That is, listeners seem to be concentrating their responses on different portions of the masculinity scale for male and female speakers. Also, when perceived as Cuencano, speakers are rated as significantly more masculine than when perceived as Lojano. These effects are independent of variant. While not significant, there is a trend seen in which the [z] guise is rated on average as more masculine/less feminine than the [s] guise.

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Table 5. Summary of best mixed-effects model for masculinity (N = 876)

	Estimate	***SE***	**t value**	***p*-value**
(Intercept)	−0.55	0.06	−9.04	<0.001
Speaker gender = male	1.10	0.06	18.47	<0.001
Perceived speaker origin = Cuenca	0.24	0.09	2.74	0.006
Perceived speaker origin = Quito	−0.08	0.08	−1.03	0.30
Perceived speaker origin = Other	−0.008	0.08	−0.10	0.92

4.5 Regional origin

A multinomial logistic regression model was fit to the ratings of perceived speaker origin in order to examine the effect of the independent variables on the choice between 'Loja,' 'Quito,' 'Cuenca,' and 'Other/I don't know,' following Chappell (2019). As is typical of multinomial logistic regression, I made four different models, each time varying the reference level of the dependent predictor (perceived speaker origin). This allows us to observe comparisons between each of the four origin categories. The first model was built with the reference level as Loja and there were no important differences between this model and the subsequent three, which had Quito, Cuenca, and 'Other' as the reference level. Thus, the first model is presented here (reference level = Loja.) Due to page width limitations, the results are split into two tables and discussed separately. Table 6 shows the first part of the results of the best model for perceived speaker origin, which includes main effects of variant and speaker gender, with an interaction between the two. In this table and the following one, positive coefficients indicate that the speakers were perceived more often as from the listed origin category (Cuenca, Quito, 'Other') than from Loja. Negative coefficients indicate the opposite – that is, speakers are heard more often as Lojano than the other origin being compared. For variant, speakers are heard significantly less frequently as Lojano in the [z] guise, and thus, are heard more as Cuencano, Quiteño, or 'Other,' in comparison to the [s] guise. For speaker gender, male speakers are heard less often as Cuencano or Quiteño than female speakers. The comparison between Loja and 'Other' is not significant.

The significant interaction between variant and speaker gender in Table 6 is better understood by examining the mosaic plots in Figure 5. In this figure, the perception of female speakers is on the left and that of male speakers is on the right. Variant is along the x-axis and the y-axis shows the percentage of responses for each of the four origin categories. These plots show that the effect of variant is much more robust for female speakers than male speakers. Female speakers are heard as less Lojana in the [z] guise, which subsequently leads to an increase in their perception as Quiteña or Cuencana. Conversely, there is not much difference

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Table 6. Summary of the best multinomial logistic regression model for perceived speaker origin, Part 1 (N = 876)

	Coefficients				*SE*				*p*-Value			
	(Int.)	Variant = [z]	Speaker = male	[z] : male	(Int.)	Variant = [z]	Speaker = male	[z] : male	(Int.)	Variant = [z]	Speaker = male	[z] : male
Perceived origin (ref. = Loja)												
Cuenca	−1.66	1.27	−0.52	−1.38	0.27	0.30	0.31	0.45	<0.001	<0.001	0.09	0.002
Quito	−0.71	0.90	−0.67	−1.002	0.21	0.26	0.25	0.37	<0.001	<0.001	0.008	0.007
Other/unknown	−1.67	0.81	0.35	−0.56	0.26	0.33	0.29	0.41	<0.001	0.01	0.22	0.18

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between the perceived origin of male speakers in the two guises, although male speakers are heard more as being from another/unknown place in the [z] guise.

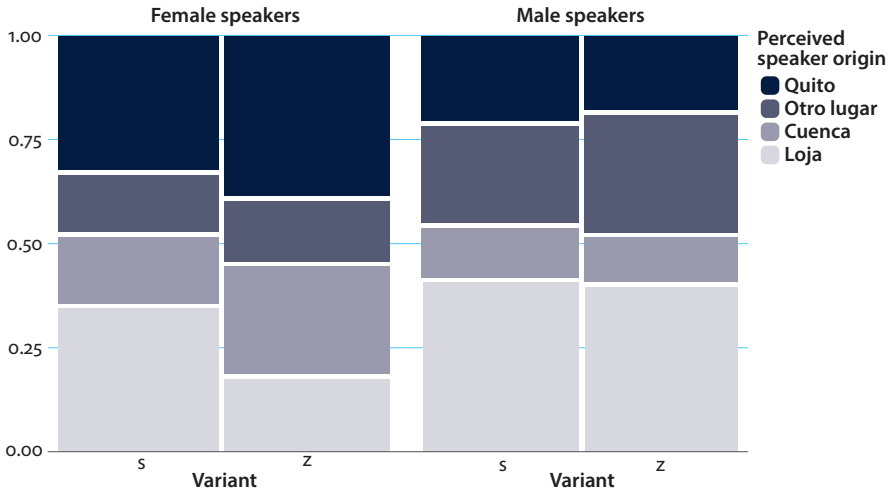

Figure 5. Mosaic plot of perceived speaker origin for [s] and [z] by speaker gender

Continuing the same model, Table 7 displays the second part of the best model for perceived speaker origin, demonstrating the effect of listener regional origin. In comparison to Lojano listeners (the reference level for "listener regional origin"), Cuencano listeners hear speakers more often as being from anywhere besides Loja, and in particular they evaluate speakers as sounding Cuencano. The positive coefficients for Quiteño listeners show a similar trend, although the only significant comparison here is in their evaluation of a speaker as Lojano versus Quiteño – favoring the latter in their evaluations. Finally, Costeño listeners evaluate speakers as sounding more Cuencano, Quiteño, or 'Other' in comparison to Lojano listeners. The interaction between listener regional origin and variant is not included in this model because it is only significant for one of the comparisons and ANOVA suggests the simpler model is a better fit. This demonstrates a trend in which listeners from different regions treat the evaluation of intervocalic [z] differently, however, it is not robust enough to be statistically significant.

The mosaic plot in Figure 6 gives a clearer picture of these results. The width of each column correlates with how many responses come from each of the four regions. The overall pattern is that listeners hear speakers as being more from their own region than from elsewhere, with the exception of Costeño listeners since 'Coast' was not one of the four options. Lojanos perceive the speakers overall as more Lojano than any other listener group. Interestingly, Cuencano and Quiteño listeners are "fooled" into thinking the speakers could be from their same place of origin, when really all the speakers are Lojano. The perception of Costeño listeners

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Table 7. Summary of the best multinomial regression model for perceived speaker origin, Part 2 (dependent variable = perceived speaker origin; independent variable = listener regional origin = 'Reg.'; N = 876)

	Coefficients				*SE*				*p*-Value			
	(Int.)	Reg. = Cuenca	Reg. = Quito	Reg. = Coast	(Int.)	Reg. = Cuenca	Reg. = Quito	Reg. = Coast	(Int.)	Reg. = Cuenca	Reg. = Quito	Reg. = Coast
Perceived origin (ref. = Loja)												
Cuenca	−1.66	2.08	0.52	1.04	0.27	0.28	0.34	0.32	0.05	<0.001	0.13	0.001
Quito	−0.71	1.30	0.87	1.01	0.21	0.25	0.25	0.26	0.22	<0.001	<0.001	<0.001
Other/unknown	−1.67	1.15	0.55	1.62	0.26	0.28	0.29	0.26	0.85	<0.001	0.06	<0.001

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is more mixed between all four categories (including many 'Other' responses), showing that perhaps their perception of the differences between the highland regions is not as nuanced as that of highlanders. This finding supports research showing that the amount of exposure to a particular dialect influences the accuracy of its identification (Clopper & Pisoni, 2004).

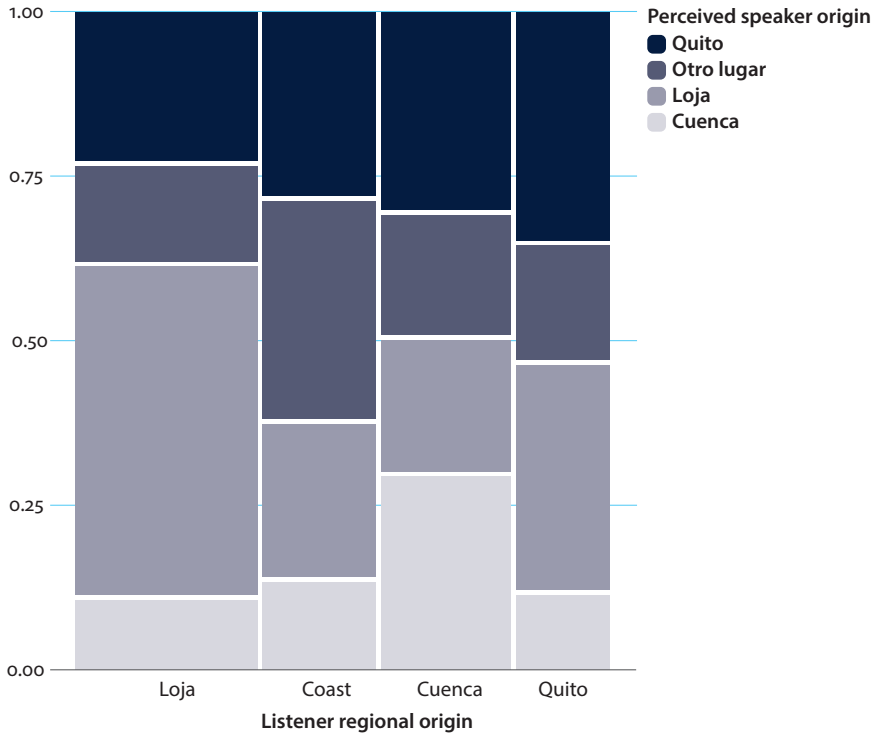

Figure 6. Mosaic plot of perceived speaker origin by listener regional origin

5. Discussion and conclusion

5.1 Interaction of gender in social perception

The perception of intervocalic /s/ voicing in HES differs greatly between male and female speakers. Intervocalic [z] is associated with lower status, less pleasant-sounding speech, and younger speakers; however, these effects are only significant for female speakers. Given that several studies have found male speakers of HES to voice more than female speakers (Strycharczuk et al., 2014; García, 2015; Schmidt, 2016; among others), it seems logical that this feature would be more "marked" for female speakers. That is, due to the preponderance of voicing in males' speech,

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whether a male speaker produces [s] or [z] does not greatly affect whether he is perceived as high or low status, pleasant or unpleasant, etc. On the other hand, since voicing is less "expected" for female speakers, it is noted as something exceptional that potentially carries information about the female speaker's status, pleasantness, age, and regional origin.

This mirrors somewhat Chappell (2016), who finds that while both male and female Costa Rican speakers are rated as lower status when producing [z], only male speakers enjoy the covert prestige that [z] affords them (more pleasant, confident, Costa Rican, and masculine). However, in the case of voicing in HES, neither male nor female speakers are perceived as more confident or pleasant when using [z] (measures of covert prestige), although regional identity is important, as discussed in the next section.

Surprisingly, despite the greater frequency with which Ecuadorians would be exposed to voicing in male speech, there is no significant difference between masculinity ratings for [s] and [z] guises. Since male speakers of HES voice more than females, one might suspect that [z] would make male speakers sound more masculine and female speakers sound less feminine, respectively. As discussed in section 4.4, this trend is seen in the data but fails to reach significance. It remains to be seen whether this trend would be borne out with more data.

5.2 Intervocalic /s/ voicing as a regional marker and potential change in progress

Apart from the social characteristics already discussed, ratings of age and perceived speaker origin are also dependent on variant. These two factors considered together can be understood under the umbrella of social and linguistic changes happening in the Ecuadorian highlands. First, for perceived speaker origin, speakers in the [z] guise are heard as significantly more Cuencano or Quiteño than in the [s] guise, and this is especially true for female speakers. Since Cuencano and Quiteño Spanish exhibit more robust, at times categorical, voicing of intervocalic /s/ (Robinson, 1979; Chappell, 2011; among others), it follows that voicing would be associated with these dialects. Voicing in Lojano Spanish, conversely, is much more variable (García, 2015). Thus, when the Lojano speakers in this experiment "put on" fully voiced intervocalic [z], it makes them sound less Lojano and more Cuencano or Quiteño. This effect also sheds light on the results for status, since Lojano Spanish is typically considered the most "conservative" of the highland dialects (see Toscano Mateus, 1953). If indeed Quiteño and Cuencano Spanish enjoy relatively less prestige than Lojano Spanish, an idea that merits further corroboration, then a feature that is associated with Quiteño and Cuencano may consequently be heard as less standard.

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Additionally, there is an effect of variant on age ratings such that female speakers sound younger in the [z] guise than the [s] guise, which parallels previous age results in production data in Lojano Spanish. García (2015) finds a significant effect of age whereby younger speakers of both genders voice more than older speakers. She is cautious to claim a change in progress given the relatively small number of speakers (N = 31 total speakers), but signals that social changes such as increased communication with other highland towns might support the idea of a change in progress. Here we see that voicing is also associated with younger generations, which gives more fuel to this argument. Voicing is associated with the neighboring highland dialects where it is found more consistently, but as it makes its way into Lojano Spanish through increased contact with these dialects, it becomes associated with the young speakers who are leading this change. Nevertheless, this is a preliminary hypothesis that needs to be further corroborated with more data. Interestingly, we see again that voicing is normal or expected for male speakers, whereas it marks youngness for female speakers.

While the results of this study suggest that intervocalic /s/ voicing is a regional marker used to distinguish someone from Quito, Cuenca, or Loja, it is important to highlight how it differs from other features. As stated previously, Gómez (2003) rightfully claims that assibilation of /r/ and delateralization of /ʎ/ serve as the major isoglosses distinguishing sub-dialects of HES. The realization of /r/ and /ɟ/-/ʎ/, however, are much more salient and commented on by naïve native speakers. There is only one mention of the "pronunciation of s's" in all of the discursive comments in the present study and naïve native speakers do not mention voicing in their descriptions of HES. Thus, intervocalic /s/ voicing contributes to the formation of highland identity, but in a more subtle way than other features. Now that it has been established as a regional marker within HES, future studies should tease apart how each feature contributes to regional identity, which could be understood under the idea of "bricolage" (Campbell-Kibler, 2010), whereby each variable adds an additional "layer" to the perception of regional identity.

One unpredicted result of the present study is that listener origin does not have a significant effect on the treatment of [s] vs. [z], even when collapsed to two, simpler categories (Coast vs. Highlands). The only place where we see an effect of listener origin is in perceived speaker origin, but this is independent of variant. While it has not been empirically studied, it is fair to assume that intervocalic /s/ voicing is not present in CES based on the fact that coastal speakers typically aspirate coda /s/, which eliminates at least one (word-final) of the three potential sites of voicing. Therefore, we might expect that coastal listeners would not have any particular social associations due to less exposure or that their social associations would differ, but instead they have the same associations that highland listeners do. It is possible that the lack of an effect of listener origin is due to the small

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number of listeners per region and that with more data, we would see more of the pattern of Walker et al. (2014), who show different variable treatment between Mexican and Puerto Rican listeners.

The differences between the results here and those of Chappell (2016) show that some of the social meanings of intervocalic /s/ voicing are local and that the indexical field (Eckert, 2008) of this variable in HES is more limited and attenuated than in Costa Rica. Chappell (2016) argues that the indexical field for intervocalic [z] includes 'uneducated,' 'low class,' 'nice,' 'confident,' 'masculine,' and 'local,' although female Costa Rican speakers are only able to access the first two of these indexes. While the indexical field of intervocalic [z] in HES does include 'low status,' it differs in the other characteristics that are included – namely, 'unpleasant,' 'young,' and 'Quiteño/Cuencano.' This confirms what Walker et al. (2014) and Pharao, Maegaard, Møller, and Kristiansen (2014) find for the indexical field of /s/ aspiration and frontedness; that is, some social meanings are global and shared between dialects, while others are local.

The results here partially uphold the theory of Chappell (2016) that female speakers may avoid nonstandard variants because they have more to lose and less to gain in doing so. Female speakers of HES are heard as lower status and less pleasant when employing intervocalic [z]. However, they are also heard as younger and more Quiteña/Cuencana, which could carry positive or negative valence depending on a myriad of contextual factors (see Giles & Billings, 2004, for studies looking at the effect of context on language attitudes). Consequently, for HES, it is not only that female speakers have more to lose and less to gain, but rather that their use of intervocalic /s/ voicing is socially-charged, whereas males' use is perceived overall as ordinary and therefore not strongly associated with any of the social characteristics examined. The potentially more complex situation of HES and the consideration of perceived speaker origin allow for an additional layer of information in the social meaning of this variable.

5.3 Conclusion

The results of this study show that intervocalic /s/ voicing in HES is associated with status, pleasantness, age, and regional origin within the highlands. However, for most of these characteristics, this social meaning is only present for female speakers. In the ratings of status, pleasantness, and age, there is no significant effect of variant ([s] vs. [z]) for the male speakers. Thus, males' use of intervocalic [z] is seen as expected and not carrying any social meaning, whereas females' use of this variant is heard as lower status, less pleasant, and younger. While no difference in social perceptions was found between listeners from different regions of Ecuador, this study does demonstrate that intervocalic /s/ voicing is a marker of

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regional identity within the highlands, such that employing intervocalic [z] makes one sound more Quiteño or Cuencano. Although all of the speakers in this study are actually from Loja, the listeners were "fooled" into thinking they were listening to Quiteños or Cuencanos at times when hearing the voiced variant. While intervocalic [z] is not associated with Lojano speech at the present time, it is not likely to stay that way as the variation takes root in Lojano Spanish.

Indeed, it is important to note that, just like the linguistic variables themselves, the language attitudes explored here are moving targets, particularly in the case of young Lojanos. My own experience and fieldwork in Loja have given me a glimpse into the ever-changing societal and linguistic landscape. Although Lojanos have always been known for being migratory (Paladines, 2006), some young Lojanos are now orienting even more towards outside of Loja for their future dreams and plans. The increased contact between Loja and other highland cities is likely the culprit for the linguistic changes that Lojano Spanish is currently undergoing, while simultaneously affecting how Lojanos view their city, at least in the middle class. Keeping a close eye on Lojano Spanish, and more broadly HES, in the future, therefore, will undoubtedly continue to be fruitful territory for examining the multifaceted interaction between language, gender, and regional identity.

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Appendix

1. Full list of stimuli; instances of intervocalic /s/ that were manipulated are bolded

Speaker M1 – male, 23 years old
***Fís**icamente **se** ha cambiado un poco, o **sea** han mejorado en cierto**s** aspectos, las carreteras.*
'It has changed physically somewhat, that is, certain aspects have improved, the highways.'
Speaker M3 – male, 29 years old
***Era una s**ociedad muy pequeña, pero …además **es** una **c**iudad muy barata.*
'It was a very small society, but …it is also a very cheap city.'
Speaker F6 – female, 47 years old
***Sí, o** sea, en lo**s** últimos años pues …ha **s**ido muy evidente.*
'Yes, that is, in the last years, well …it has been very evident.'
Speaker F12 – female, 20 years old
***Por es**o me encanta, yo quiero ir a un país así, hay más oportunidades.*
'That's why I love it, I want to go to a country like that, there are more opportunities.'

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2. Experimental lists – stimuli order

List 1	**List 2**
1. F6 [s]	1. F6 [z]
2. M3 [z]	2. M3 [s]
3. *filler*	3. *filler*
4. F12 [z]	4. *filler*
5. *filler*	5. F12 [s]
6. *filler*	6. *filler*
7. M1 [s]	7. *filler*
8. *filler*	8. M1 [z]
9. F6 [z]	9. F6 [s]
10. M3 [s]	10. M3 [z]
11. F12 [s]	11. *filler*
12. *filler*	12. F12 [z]
13. M1 [z]	13. *filler*
14. *filler*	14. M1 [s]
15. *filler*	15. *filler*
16. *filler*	16. *filler*

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